

EXPERIMENT 20: Prolog Program for Planets Database

Aim

To write and implement a **Prolog program to create a database of planets** and retrieve information about planets using facts and rules.

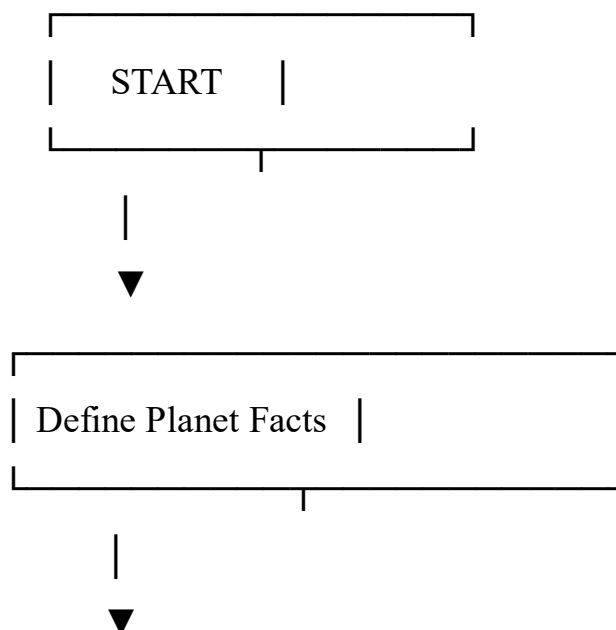
Objective

- To understand facts and rules in Prolog.
- To store information about planets.
- To retrieve planet details using queries.
- To determine properties such as planet type and position.

Algorithm

5. Start the program.
6. Define facts for planets in the Solar System.
7. Store information such as planet name, position, and type.
8. Define rules to identify terrestrial and gas-giant planets.
9. Execute queries to retrieve planet information.
10. Display the result.
11. Stop the program.

Flowchart



Define Planet Rules



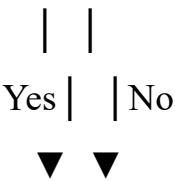
Enter a Query



Search Database



Found?



Display Result

Display "Not Found"



```
┌  
|  END  |  
└
```

Code

% Prolog Program for Planets Database

% Planet facts

planet(mercury).

planet(venus).

planet(earth).

planet(mars).

planet(jupiter).

planet(saturn).

planet(uranus).

planet(neptune).

% Position of planets from the Sun

position(mercury, 1).

position(venus, 2).

position(earth, 3).

position(mars, 4).

position(jupiter, 5).

position(saturn, 6).

position(uranus, 7).

position(neptune, 8).

% Planet types

type(mercury, terrestrial).

type(venus, terrestrial).

type(earth, terrestrial).

type(mars, terrestrial).

type(jupiter, gas_giant).

type(saturn, gas_giant).

type(uranus, ice_giant).

type(neptune, ice_giant).

% Rule to find terrestrial planets

terrestrial_planet(Planet) :-

type(Planet, terrestrial).

% Rule to find giant planets

giant_planet(Planet) :-

type(Planet, gas_giant).

% Rule to find the position of a planet

planet_position(Planet, Position) :-

position(Planet, Position).

Queries

1. Find all planets

?- planet(X).

Output

X = mercury ;

X = venus ;

X = earth ;

X = mars ;

X = jupiter ;

X = saturn ;

X = uranus ;

X = neptune.

2. Find the position of Earth

?- planet_position(earth, X).

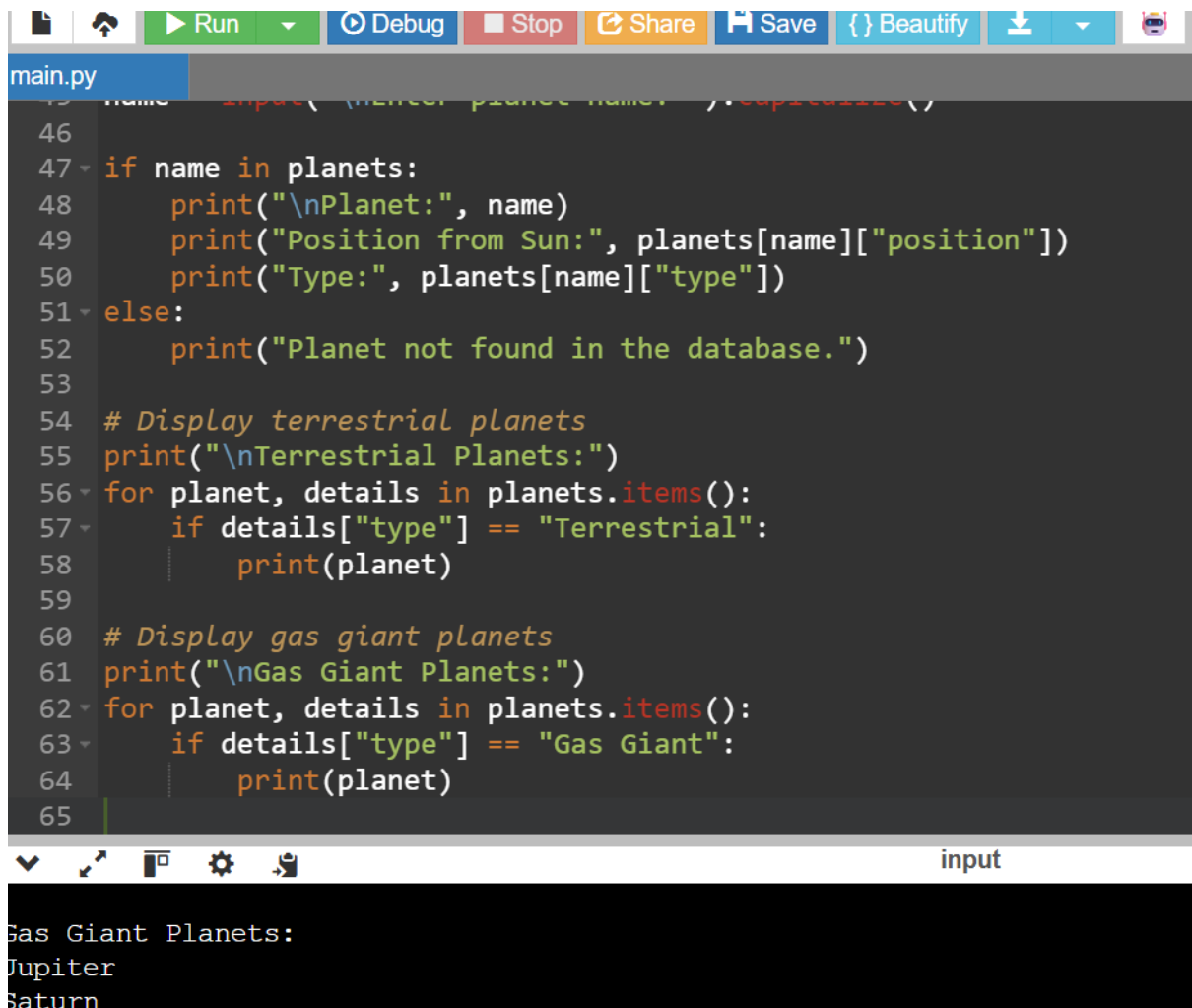
Output

X = 3.

3. Find terrestrial planets

?- terrestrial_planet(X).

Output



```
45 name = input( "Enter planet name: ").capitalize()
46
47 if name in planets:
48     print("\nPlanet:", name)
49     print("Position from Sun:", planets[name]["position"])
50     print("Type:", planets[name]["type"])
51 else:
52     print("Planet not found in the database.")
53
54 # Display terrestrial planets
55 print("\nTerrestrial Planets:")
56 for planet, details in planets.items():
57     if details["type"] == "Terrestrial":
58         print(planet)
59
60 # Display gas giant planets
61 print("\nGas Giant Planets:")
62 for planet, details in planets.items():
63     if details["type"] == "Gas Giant":
64         print(planet)
65
```

input

Gas Giant Planets:
Jupiter
Saturn

Result

The Prolog program was successfully implemented to create a **Planets Database** and retrieve information about planets using facts and rules.

Conclusion

The program demonstrates how **Prolog facts, rules, variables, and queries** can be used to create and search a simple knowledge database containing information about planets.